Hope
IN OUR
Suffering

Romans 5:3-4 (NIV)

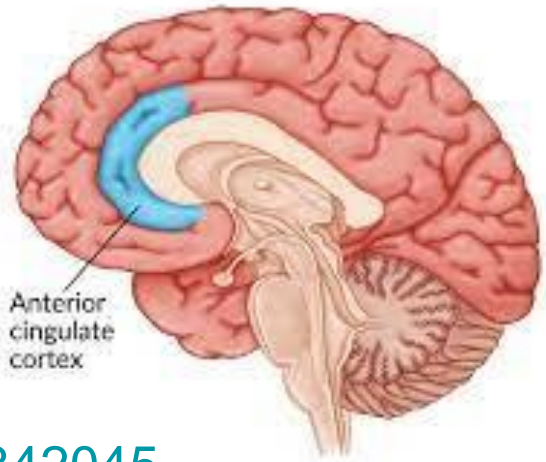
“Not only so, but we also **glory in our sufferings**, because we know that **suffering produces perseverance**; perseverance, character; and character, hope.”

Suffering > Perseverance > Character > Hope

Cingulate
Cortex

Anterior Cingulate Cortex (ACC)

"Overall, anterior cingulate cortex appears to play a crucial role in initiation, motivation, and goal-directed behaviours"



Sources:
https://academic.oup.com/brain/article/118/1/279/342045

Anterior MidCingulate Cortex (aMCC)

"its position as a structural and functional hub allows the aMCC to integrate signals from diverse brain systems to predict energy requirements that are needed for attention allocation, encoding of new information, and physical movement, all in the service of goal attainment."

Sources:
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7381101/
https://www.sciencedirect.com/science/article/pii/S0010945219303326

Anterior MidCingulate Cortex (aMCC)

- ‘Super Agers’ have a larger aMCC!
- aMCC is closely linked to willpower (I will or I will not!).
- Smaller and less aMCC usage tied to depression.

Sources:
[Dr. Andrew Huberman - Huberman Lab Podcast]

Sufferings (Romans 5:3-4)

Word: qliyij Pronounce: thlip'-sis Strongs Number: G2347

Orig: from 2346; pressure (literally or figuratively):--afflicted(-tion), anguish, burdened, persecution, tribulation, trouble. G2346

1) a pressing, pressing together, pressure

2) metaph. oppression, affliction, tribulation, distress, straits

Suffering Affects The aMCC

When you do things you don’t like, the aMCC grows!

- Successful dieters…
- People who workout more…

Source:
https://www.pnas.org/doi/full/10.1073/pnas.1313475110
Dr. Andrew Huberman - Huberman Lab Podcast
https://roybaumeister.com

Allostasis

"We present evidence that the aMCC, positioned at the intersection of multiple brain networks, is **wired to integrate signals relating to allostasis** with its sensory consequences, termed interoception, as well as with cognitive control processes, sensory, and motor functions."

"Allostasis is defined as the process of maintaining homeostasis through the adaptive change of the organism's internal environment to **meet perceived and anticipated demands**"

Sources:
https://www.sciencedirect.com/topics/immunology-and-microbiology/allostasis
https://www.sciencedirect.com/science/article/pii/S2215091918300208

Perseverance (Romans 5:3-4)

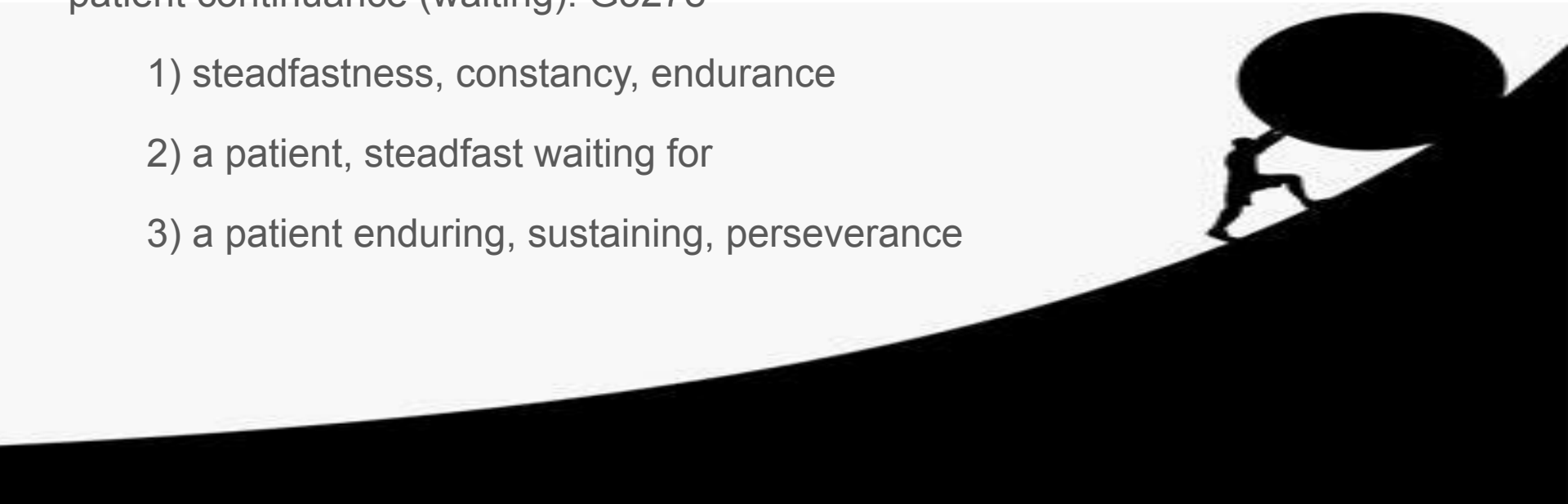
Word: upomonh Pronounce: hoop-om-on-ay' Strongs Number: G5281

Orig: from 5278; cheerful (or hopeful) endurance, constancy:--enduring, patience, patient continuance (waiting). G5278

1) steadfastness, constancy, endurance

2) a patient, steadfast waiting for

3) a patient enduring, sustaining, perseverance



Tenacity

"Tenacity–**persistence in the face of challenge**–has received increasing attention, particularly because it contributes to better academic achievement, career opportunities and health outcomes."

- aMCC directly communicates with the brain and body to modulate Tenacity!
- aMCC gives us the sense to resist, move, and act!

Sources:
Dr. Andrew Huberman - Huberman Lab Podcast
https://www.sciencedirect.com/science/article/pii/S0010945219303326

Character (Romans 5:3-4)

Dictionary: “The mental and moral qualities distinctive to an individual.”

Word: dokimh Pronounce: dok-ee-may' Strongs Number: G1382

Orig: from the same as 1384; test (abstractly or concretely); by implication, trustiness:--experience(-riment), proof, trial. G1384

1) proving, trial

2) approved, tried character

3) a proof, a specimen of tried worth

The Will To Live

- “Scientists are starting to think of the aMCC not just as the seat of willpower, but perhaps as **the seat of the will to live**.”

Source:
[“David Goggins: How to Build Immense Inner Strength” - Huberman Lab Podcast]

Hope (Romans 5:3-4)

Dictionary: “A feeling of expectation and desire for a certain thing to happen.”

Word: elpij Pronounce: el-pece' Strongs Number: G1680

Orig: from a primary elpo (to anticipate, usually with pleasure); expectation (abstractly or concretely) or confidence:--faith, hope.

1) expectation of evil, fear

2) expectation of good, hope

3) on hope, in hope, having hope

“Glory in our sufferings”

- The aMCC grows through acts we do not like to do!
- We can build the aMCC up.
- We can break the aMCC down.
- It is a continual process.

Source:
“David Goggins: How to Build Immense Inner Strength” - Huberman Lab Podcast

Romans 5:3-4 (NIV)

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Suffering > Perseverance > Character > Hope